Curriculum Vitae

Tom de Geus

"It's not the answers you give, but the questions you ask" – Voltaire

Personal information

Dr.ir. Tom de Geus
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tom@geus.me, +41 77 993 72 35, www.geus.me
Dutch (mother tongue), English (fluent), French (fluent), German (excellent)
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Employment (academic positions only)

1. Patent expert & examiner. 2024–pres.
   Swiss Federal Institute of Intellectual Property, CH.
2. Ambizione fellow (see fellowships). 2019–2024
   Institute of Physics, École Polytechnique Fédérale de Lausanne (EPFL), CH.
3. Rubicon post-doc fellow (see fellowships). 2016–2019
   Director: Prof.dr. Matthieu Wyart.
   Institute of Physics, École Polytechnique Fédérale de Lausanne (EPFL), CH.
4. Post-doc (valorisation project for M2i and TATA Steel Europe). 2016
   Director: Prof.dr.ir. Marc Geers.
   Department of Mechanical Engineering, Eindhoven University of Technology, NL.
5. PhD researcher. 2012–2016
   Directors: Prof.dr.ir. Marc Geers, dr.ir. Ron Peerlings.
   Department of Mechanical Engineering, Eindhoven University of Technology, NL.
   Materials Innovation Institute (M2i), Delft, NL.

Education

1. PhD Mechanical Engineering (*cum laude*). Defended: 03.05.2016. 2012–2016
   Promotor: Prof.dr.ir. Marc Geers, co-promotor: Dr.ir. Ron Peerlings.
   Eindhoven University of Technology (TU/e), The Netherlands.
2. Graduate School Engineering Mechanics. 2012–2015
   Supported by 3TU (Joint Universities of Technology in The Netherlands).
3. Master Mechanical Engineering (*with great appreciation*). 2009–2012
   Thesis directors: Prof.dr.ir. Marc Geers, dr.ir. Ron Peerlings.
   Eindhoven University of Technology, The Netherlands.
4. Internship (*excellent evaluation*). 2010
   Director: Prof.dr. Katia Bertoldi.
   Harvard University, School of Engineering and Applied Sciences, Cambridge, USA.
5. Bachelor Mechanical Engineering (*with great appreciation*). 2004–2009
   Minor: entrepreneurship.
   Eindhoven University of Technology, The Netherlands.
6. Pre-university secondary education. 1997–2004
   Specialisation: science and technology.
   Lorentz Casimir Lyceum, Eindhoven, The Netherlands.

Fellowships, Prizes, Awards

Fellowships

1. Ambizione fellowship, Grant No. PZ00P2_185843 — 2019–2024
   Swiss National Science Foundation (FNSF). Grant: 567 kCHF.
2. Rubicon fellowship, Grant No. 680-50-1520 — 2016–2019
   The Netherlands Organisation for Scientific Research (NWO). Grant: 160 k€.

Prizes

1. Martinus van Marum award for best PhD thesis in the engineering sciences in a period of five years, Koninklijke Hollandsche Maatschappij der Wetenschappen (Royal Holland Society of Sciences and Humanities). — 2017
   Prize: acceptance colloquium, a medal, and 12500 €.
2. Best PhD thesis in Mechanical Engineering of the calendar year 2016, Eindhoven University of Technology (TU/e). Prize: 250 €. — 2017

Awards

1. Winner of the 7th ECCOMAS PhD Olympiad. — 2017
   Award: 500 €.
2. Poster award for best innovative research value, — 2016
   M2i annual research conference. Award: 250 €.
3. Poster award, — 2013
   M2i annual research conference. Award: 250 €.
4. Entrepreneurship award for best business case, — 2008
   Eindhoven University of Technology. Award: 1000 €.

Publications (chronological)

[29] H. Chen et al. "Investigating criticality in rock failure using fiber-optic strain sensing". In: *Geophys. Res. Lett.* 52.14 (2025), e2025GL115010. DOI: 10.1029/2025GL115010.

[28] T.W.J. de Geus, A. Rosso, and M. Wyart. "Dynamical Heterogeneities of Thermal Creep in Pinned Interfaces". In: *arXiv preprint: 2401.09830* arXiv:2401.09830 (2024). DOI: 10.48550/arXiv.2401.09830.

[27] T.W.J. de Geus and M. Wyart. "Short-Range Depinning in the Presence of Velocity-Weakening". In: *arXiv preprint: 2411.06732* arXiv:2411.06732 (2024). DOI: 10.48550/arXiv.2411.06732.

[26] E. El Sergany, M. Wyart, and T.W.J. de Geus. "Armouring of a Frictional Interface by Mechanical Noise". In: *J Stat Phys* 191.10 (2024), p. 134. DOI: 10.1007/s10955-024-03339-z. arXiv: 2301.13802.

[25] S. Poincloux, P.M. Reis, and T.W.J. De Geus. "Stick-Slip in a Stack: How Slip Dissonance Reveals Aging". In: *Phys. Rev. Research* 6.1 (2024), p. 013080. DOI: 10.1103/PhysRevResearch.6.013080. arXiv: 2301.13745.

[24] T.W.J. de Geus. "E-L-M, Simple Principles to Keep Data and Code Alive". In: *Prepint* (2023). DOI: 10.31222/osf.io/8tzb9.

[23] T.W.J. de Geus and M. Wyart. "Scaling Theory for the Statistics of Slip at Frictional Interfaces". In: *Phys. Rev. E* 106.6 (2022), p. 065001. DOI: 10.1103/PhysRevE.106.065001. arXiv: 2204.02795.

[22] W. Ji et al. "Mean-Field Description for the Architecture of Low-Energy Excitations in Glasses". In: *Phys. Rev. E* 105.4 (2022), p. 044601. DOI: 10.1103/PhysRevE.105.044601. arXiv: 2106.13153.

[21] M. Popović et al. "Scaling Description of Creep Flow in Amorphous Solids". In: *Phys. Rev. Lett.* 129.20 (2022), p. 208001. DOI: 10.1103/PhysRevLett.129.208001. arXiv: 2111.04061.

[20] M. Popović et al. "Thermally Activated Flow in Models of Amorphous Solids". In: *Phys. Rev. E* 104.2 (2021), p. 025010. DOI: 10.1103/PhysRevE.104.025010. arXiv: 2009.04963.

[19] W. Ji et al. "Thermal Origin of Quasilocalized Excitations in Glasses". In: *Phys. Rev. E* 102.6 (2020), p. 062110. DOI: 10.1103/PhysRevE.102.062110. arXiv: 1912.10537.

[18] J.C. Volmer, T.W.J. de Geus, and R.H.J. Peerlings. "Improving the Initial Guess for the Newton-Raphson Protocol in Time-Dependent Simulations". In: *J. Comput. Phys.* 420 (2020), p. 109721. DOI: 10.1016/j.jcp.2020.109721. arXiv: 1912.12140.

[17] J. Vondřejc and T.W.J. de Geus. "Energy-Based Comparison between the Fourier–Galerkin Method and the Finite Element Method". In: *J. Comput. Appl. Math.* 374 (2020), p. 112585. DOI: 10.1016/j.cam.2019.112585. arXiv: 1709.08477.

[16] T.W.J. de Geus et al. "How Collective Asperity Detachments Nucleate Slip at Frictional Interfaces". In: *Proc. Natl. Acad. Sci.* 116.48 (2019), pp. 23977–23983. DOI: 10.1073/pnas.1906551116. arXiv: 1904.07635.

[15] W. Ji et al. "Theory for the Density of Interacting Quasilocalized Modes in Amorphous Solids". In: *Phys. Rev. E* 99.2 (2019), p. 023003. DOI: 10.1103/PhysRevE.99.023003. arXiv: 1806.01561.

[14] M. Popović, T.W.J. de Geus, and M. Wyart. "Elastoplastic Description of Sudden Failure in Athermal Amorphous Materials during Quasistatic Loading". In: *Phys. Rev. E* 98.4 (2018), p. 040901. DOI: 10.1103/PhysRevE.98.040901. arXiv: 1803.11504.

[13] T.W.J. de Geus, R.H.J. Peerlings, and M.G.D. Geers. "Fracture in Multi-Phase Materials: Why Some Microstructures Are More Critical than Others". In: *Eng. Fract. Mech.* 169 (2017), pp. 354–370. DOI: 10.1016/j.engfracmech.2016.08.009. arXiv: 1603.08910.

[12] T.W.J. de Geus et al. "Finite Strain FFT-based Non-Linear Solvers Made Simple". In: *Comput. Methods Appl. Mech. Eng.* 318 (2017), pp. 412–430. DOI: 10.1016/j.cma.2016.12.032. arXiv: 1603.08893.

[11] J. Zeman et al. "A Finite Element Perspective on Nonlinear FFT-based Micromechanical Simulations". In: *Int. J. Numer. Methods Eng.* 111.10 (2017), pp. 903–926. DOI: 10.1002/nme.5481. arXiv: 1601.05970.

[10] T.W.J. de Geus, R.H.J. Peerlings, and M.G.D. Geers. "Competing Damage Mechanisms in a Two-Phase Microstructure: How Microstructure and Loading Conditions Determine the Onset of Fracture". In: *Int. J. Solids Struct.* 97–98 (2016), pp. 687–698. DOI: 10.1016/j.ijsolstr.2016.03.029. arXiv: 1603.05841.

[9] T.W.J. de Geus et al. "Fracture Initiation in Multi-Phase Materials: A Statistical Characterization of Microstructural Damage Sites". In: *Mater. Sci. Eng. A* 673 (2016), pp. 551–556. DOI: 10.1016/j.msea.2016.06.082. arXiv: 1603.08898.

[8] T.W.J. de Geus et al. "Fracture Initiation in Multi-Phase Materials: A Systematic Three-Dimensional Approach Using a FFT-based Solver". In: *Mech. Mater.* 97 (2016), pp. 199–211. DOI: 10.1016/j.mechmat.2016.02.006. arXiv: 1604.03817.

[7] T.W.J. de Geus et al. "Microscopic Plasticity and Damage in Two-Phase Steels: On the Competing Role of Crystallography and Phase Contrast". In: *Mech. Mater.* 101 (2016), pp. 147–159. DOI: 10.1016/j.mechmat.2016.07.014. arXiv: 1603.05847.

[6] T.W.J. de Geus et al. "Systematic and Objective Identification of the Microstructure around Damage Directly from Images". In: *Scr. Mater.* 113 (2016), pp. 101–105. DOI: 10.1016/j.scriptamat.2015.10.007. arXiv: 1604.03814.

[5] J. van Beeck et al. "Predicting Deformation-Induced Polymer–Steel Interface Roughening and Failure". In: *Eur. J. Mech. - ASolids* 55 (2016), pp. 1–11. DOI: 10.1016/j.euromechsol.2015.08.002.

[4] T.W.J. de Geus, R.H.J. Peerlings, and M.G.D. Geers. "Microstructural Modeling of Ductile Fracture Initiation in Multi-Phase Materials". In: *Eng. Fract. Mech.* 147 (2015), pp. 318–330. DOI: 10.1016/j.engfracmech.2015.04.010. arXiv: 1604.03811.

[3] T.W.J. de Geus, R.H.J. Peerlings, and M.G.D. Geers. "Microstructural Topology Effects on the Onset of Ductile Failure in Multi-Phase Materials - A Systematic Computational Approach". In: *Int. J. Solids Struct.* 67–68 (2015), pp. 326–339. DOI: 10.1016/j.ijsolstr.2015.04.035. arXiv: 1604.02858.

[2] T.W.J. de Geus, R.H.J. Peerlings, and M.G.D. Geers. "Topology and Morphology Influences on the Onset of Ductile Failure in a Two-phase Microstructure". In: *Procedia Mater. Sci.* 3 (2014), pp. 598–603. DOI: 10.1016/j.mspro.2014.06.099. arXiv: 1604.03258.

[1] T.W.J. de Geus, R.H.J. Peerlings, and C.B. Hirschberger. "An Analysis of the Pile-up of Infinite Periodic Walls of Edge Dislocations". In: *Mech. Res. Commun.* 54 (2013), pp. 7–13. DOI: 10.1016/j.mechrescom.2013.08.010. arXiv: 1604.02848.

Conferences (invited oral)

[10] *A Theory for the Statistics of Slip at a Frictional Interface: Unifying Rate-and-State and Depinning Approaches*. Ascona, Switzerland: Friction and wear across scales, Aug. 16, 2022.

[9] *Scaling Theory for the Nucleation of Slip at the Frictional Interface*. Njord Center, Oslo, Norway: Friction and faulting workshop, June 10, 2022.

[8] *Critical Flow Properties of a Frictional Interface*. JMC17. France: 17èmes journées de la matière condensée, Aug. 24, 2021.

[7] *Does Inertia Cause Stick-Slip Friction?* Oslo, Norway: EarthFlows meeting: Complexity in solid earth and geophysical flows, June 19, 2019.

[6] *How Is Slip Nucleated at a Frictional Interface?* SES 2019. Saint Louis, United States: 56th. Annual technical meeting of the Society of Engineering Science, Oct. 13, 2019.

[5] *Nucleation of Slip at the Frictional Interface (Approximative Title)*. Madrid, Spain: 55th annual technical meeting of the Society of Engineering Science (SES), Oct. 10, 2018.

[4] *Fracture in Multi-Phase Materials: Why Some Microstructures Are More Critical than Others*. CFRAC 2017. Nantes, France: Fifth International Conference on Computational Modeling of Fracture and Failure of Materials and Structures, June 14, 2017.

[3] *Fracture Nucleation in Multi-Phase Metals (Approximative Title)*. YIC 2019. Milan, Italy: Seventh ECCOMAS PhD Olympiad, IV ECCOMAS Young Investigators Conference, Sept. 3, 2017.

[2] *Nulceation of Fracture in a DP-steel (Approximative Title)*. ESMC9. Madrid, Spain: 9th European Solid Mechanics Conference, July 5, 2015.

[1] *Unraveling the Three-Dimensional Fracture Mechanisms of a Multi-Phase Material*. COMPLAS XIII. Barcelona, Spain: XIII International Conference on Computational Plasticity., Sept. 1, 2015.

Conferences (oral)

[25] *Criticality and Nucleation of Slip at the Frictional Interface*. SSD 31. Fribourg, Switzerland: Swiss Soft Days, Apr. 14, 2023.

[24] *Criticality and Nucleation of Slip at the Frictional Interface*. Les Houches, France: Interaction, disorder, elasticity, Apr. 3, 2023.

[23] *A Theory for the Statistics of Slip at a Frictional Interface: Unifying Rate-and-State and Depinning Approaches*. SPS. Fribourg, Switzerland: Swiss Physical Society, annual meeting, June 27, 2022.

[22] *A Theory for the Statistics of Slip at a Frictional Interface: Unifying Rate-and-State and Depinning Approaches*. CMIS2022. Lausanne, Switzerland: Contact Mechanics International Symposium 2022, Apr. 24, 2022.

[21] *A Theory for the Statistics of Slip at a Frictional Interface: Unifying Rate-and-State and Depinning Approaches*. SSD 29. Villigen, Switzerland: Swiss Soft Days, Apr. 5, 2022.

[20] *Accelerated Ageing Multi-Fault Systems?* Oslo, Norway: ECCOMAS, June 1, 2022.

[19] *Scaling Theory for the Nucleation of Slip at the Frictional Interface*. USNCTAM2022. Austin, USA: 19th U.S. national congress on theoretical and applied mechanics, June 21, 2022.

[18] *Nucleation of Slip at the Frictional Interface: Avalanches or Fracture?* Spetses, Greece: Disordered elastic systems, Sept. 6, 2021.

[17] *Does Inertia Induce Stick-Slip Friction?* Les Houches, France: Avalanche dynamics and precursors of catastrophic events, Feb. 4, 2019.

[16] *How Is Slip Nucleated at a Frictional Interface?* ICNEM 19. Kraków, Poland: 24th international conference on nonlinear elasticity in materials, June 23, 2019.

[15] *FFT-based Solver (in Finite Strain) Made Simple.* COMPLAS XIV. Barcelona, Spain: XIV International Conference on Computational Plasticity., Sept. 5, 2017.

[14] *FFT-based Solution: A Complex Method Made Simple (Even for 3-D Finite Strain).* Crete, Greece: ECCOMAS 2016, June 5, 2016.

[13] *Fourier-Galerkin Methods as Alternative to Finite Element Method for Numerical Homogeniaation.* Crete, Greece: ECCOMAS 2016, June 5, 2016.

[12] *Nulceation of Fracture in a DP-steel (Approximative Title).* ITCAM2016. Montreal, Canada: 24th International Congress of Theoretical and Applied Mechanics, Aug. 22, 2016.

[11] *Nulceation of Fracture in a DP-steel (Approximative Title).* ECF21. Catania, Italy: 21st European Conference on Fracture, June 20, 2016.

[10] *A Systematic Micromechanical Analysis of the Onset of Fracture in Multi-Phase Materials.* Houffalize, Belgium: Euromech colloquium 570: impact of microstructure on plasticity, Oct. 20, 2015.

[9] *Nulceation of Fracture in a DP-steel (Approximative Title).* Sint-Michielsgestel, The Netherlands: Materials innovation institute (M2i) 16th annual conference, Dec. 7, 2015.

[8] *Nulceation of Fracture in a DP-steel (Approximative Title).* USNCCM13. San Diego, United States: 13th U.S. National Congress on Computational Mechanics, July 27, 2015.

[7] *Nulceation of Fracture in a DP-steel (Approximative Title).* EMMC14. Gothenburg, Sweden: 4th European Mechanics of Materials Conference, Aug. 27, 2014.

[6] *Nulceation of Fracture in a DP-steel (Approximative Title).* WCCM2014. Barcelona, Spain: World Congress on Computational Mechanics, June 20, 2014.

[5] *Topology and Morphology Influences on the Onset of Ductile Failure in a Two- Phase Microstructure.* ECF20. Trondheim, Norway: 20th European Conference on Fracture, June 30, 2014.

[4] *Nulceation of Fracture in a DP-steel (Approximative Title).* Lunteren, The Netherlands: Engineering Mechanics Symposium, Oct. 2013.

[3] *Nulceation of Fracture in a DP-steel (Approximative Title).* COMPLAS XII. Barcelona, Spain: XII International Conference on Computational Plasticity, Sept. 3, 2013.

[2] *On the Effect of Microstructural Morphology on Ductile Failure in Multi-Phase Materials.* ICF13. Beijing, China: 13th International Conference on Fracture 2013, June 16, 2013.

[1] *Pile-up of Edge Dislocations (Approximative Title).* SRC. Eindhoven, The Netherlands: Student research conference, 2012.

Conferences (posters)

[15] T.W.J. de Geus. *Nucleation of Slip at the Frictional Interface (Approximate Title).* Disorder's Role in Glass Formation and Deformation. Leiden, The Netherlands: Lorentz Center, July 15, 2022.

[14] A. Devresse et al. *Relating Microstructural Morphology and Mechanics to the Onset of Fracture in Multi-Phase Materials.* PASC22. Basel, Switzerland: Platform for Advanced Scientific Computing, 2022-6-27, 2022.

[13] T.W.J. de Geus. *Nucleation of Slip at the Frictional Interface (Approximate Title).* MEPHISTO. Cargèse, France: MEchanics and PHysics of STretchable Objects, Aug. 7, 2018.

[12] T.W.J. de Geus. *Nucleation of Slip at the Frictional Interface (Approximate Title).* Physics Day. Lausanne, Switzerland: EPFL, 2018.

[11] T.W.J. de Geus. *Nucleation of Slip at the Frictional Interface (Approximate Title).* Micro-Nano Models for Tribology. Leiden, The Netherlands: Lorentz Center, Jan. 20, 2017.

[10] T.W.J. de Geus. *Fracture Nucleation in DP Steel (Approximate Title)*. Noordwijk, The Netherlands: Annual M2i Symposium, 2016.

[9] T.W.J. de Geus. *Fracture Nucleation in DP Steel (Approximate Title)*. Lunteren, The Netherlands: Annual Engineering Mechanics Symposium, 2016.

[8] T.W.J. de Geus. *Fracture Nucleation in DP Steel (Approximate Title)*. Noordwijk, The Netherlands: Annual M2i Symposium, 2015.

[7] T.W.J. de Geus. *Fracture Nucleation in DP Steel (Approximate Title)*. Lunteren, The Netherlands: Annual Engineering Mechanics Symposium, 2015.

[6] T.W.J. de Geus. *Relating Microstructural Morphology and Mechanics to the Onset of Fracture in Multi-Phase Materials*. Eindhoven, The Netherlands: Euromech colloquium 559: multi-scale computational methods, Feb. 25, 2015.

[5] T.W.J. de Geus. *Fracture Nucleation in DP Steel (Approximate Title)*. Noordwijk, The Netherlands: Annual M2i Symposium, 2014.

[4] T.W.J. de Geus. *Fracture Nucleation in DP Steel (Approximate Title)*. Lunteren, The Netherlands: Annual Engineering Mechanics Symposium, 2014.

[3] T.W.J. de Geus. *Fracture Nucleation in DP Steel (Approximate Title)*. Noordwijk, The Netherlands: Annual M2i Symposium, 2013.

[2] T.W.J. de Geus. *Fracture Nucleation in DP Steel (Approximate Title)*. Lunteren, The Netherlands: Annual Engineering Mechanics Symposium, 2013.

[1] T.W.J. de Geus. *Fracture Nucleation in DP Steel (Approximate Title)*. Lunteren, The Netherlands: Annual Engineering Mechanics Symposium, 2012.

Colloquia (invited)

[20] *Predicting the Unpredictable*. Thun, Switzerland: EMPA, Jan. 17, 2024.

[19] *Slippery Disordered Solids*. Villingen, Switzerland: PSI, Nov. 7, 2023.

[18] *Slippery Disordered Solids*. Thun, Switzerland: EMPA, Oct. 31, 2023.

[17] *Creep Flow in Amorphous Solids (Approximate Title)*. Fribourg, Switzerland: University of Fribourg, May 16, 2022.

[16] *Slip Nucleation at the Frictional Interface (Approximate Title)*. Grenoble, France: Université Grenoble Alpes, Nov. 9, 2022.

[15] *Slip Nucleation at the Frictional Interface (Approximate Title)*. Fribourg, Switzerland: University of Fribourg, Nov. 3, 2022.

[14] *Slip Nucleation at the Frictional Interface (Approximate Title)*. Zürich, Switzerland: Universität Zürich, Oct. 4, 2022.

[13] *Slip Nucleation at the Frictional Interface (Approximate Title)*. LiPhy. Grenoble, France: Université Grenoble Alpes, Sept. 19, 2022.

[12] *Slip Nucleation at the Frictional Interface (Approximate Title)*. Zürich, Switzerland: ETH Zürich, Sept. 7, 2022.

[11] *Slip Nucleation at the Frictional Interface (Approximate Title)*. Eindhoven, The Netherlands: Eindhoven University of Technology, June 17, 2022.

[10] *Slip Nucleation at the Frictional Interface (Approximate Title)*. Lausanne, Switzerland: University of Lausanne, June 15, 2022.

[9] *Slip Nucleation at the Frictional Interface*. Lausanne, Switzerland: EPFL, Apr. 21, 2021.

[8] *Slip Nucleation at the Frictional Interface (Approximate Title)*. Lyon, France: ENS lyon, Nov. 9, 2021.

[7] *Slip Nucleation at the Frictional Interface*. Amsterdam, The Netherlands: University of Amsterdam, Dec. 21, 2020.

[6] *Nucleation of Slip at the Frictional Interface*. Amsterdam, The Netherlands: ARCNL, June 12, 2019.

[5] *Slip Nucleation at the Frictional Interface (Approximate Title)*. Prague, Czech Republic: Czech Technical University, Jan. 30, 2019.

[4] *Fracture Nucleation in Multi-Phase Materials (Approximate Title)*. Eindhoven, The Netherlands: Eindhoven University of Technology, Feb. 12, 2018.

[3] *Fracture Nucleation in Multi-Phase Materials (Approximate Title)*. Paris, France: MINES ParisTech, Sept. 13, 2017.

[2] *Fracture Nucleation in Multi-Phase Materials (Approximate Title)*. cemef. Sophia-Antipolis, France: MINES ParisTech, Nov. 4, 2016.

[1] *Fracture Nucleation in Multi-Phase Materials (Approximate Title)*. IJmuiden, The Netherlands: TATA Steel Europe, Apr. 2016.

(Co-)Supervised junior researchers: PhD, master, bachelor (chronological)

[32] S. Mortgat. "Emporal Correlations in the Creep Regime". Semester Project. EPFL, 2024.

[31] T. Pottie. "Fractal Dimension of Avalanches in Amorphous Solids". Semester Project. EPFL, 2024.

[30] L. Bugnard. "Modelling Fractal Disorder in an Amorphous Solid". Double Semester Project. EPFL, 2023.

[29] E. El Sergany. "Measuring P(x) Experimentally in a Frictional Interface". MSc Thesis. EPFL, 2023.

[28] T. Fjellman. "Measuring the Effect of Temperature in a Burridge-Knopoff Model with Inertia". Double Semester Project. EPFL, 2023.

[27] S. Mortgat. "Spatio-Temporal Correlations in Thermal Avalanches in Depinning". Semester Project. EPFL, 2023.

[26] T. Pottie. "Thermal Avalanches in Amorphous Solids". Semester Project. EPFL, 2023.

[25] E. El Sergany. "A Simple Model for P(x) in a Prandtl-Tomlinson Model". Semester Project. EPFL, 2022.

[24] T. Fjellman. "Measuring the Effect of Temperature in a Burridge-Knopoff Model with Inertia". Summer Internship. EPFL, 2022.

[23] Y. Li. "Implementing Temperature in an Inertial System". Semester Project. EPFL, 2022.

[22] L. Linder. "Looking for the Origin of Rate-and-State Friction". Semester Project. EPFL, 2022.

[21] W. Ji. "Local Excitations in Amorphous Solids". PhD thesis. EPFL, 2021.

[20] T. Salomon. "Finite Size Effect in the Hébraud-Lequeux Model". Semester Project. EPFL, 2021.

[19] C. Georgantas. "Implementation of a Cylindrical Element in GooseFEM". Semester Project. EPFL, 2019.

[18] F. Chappuis. "Capillary Instabilities in Liquid and Solid Cylinders". Semester Project. EPFL, 2018.

[17] T. Ma. "A Microscopic Study of Surface Adhesion in Friction Problem". Double Semester Project. EPFL, 2018.

[16] C. Ylla Arbós. "Modeling of Inflatable Structures Using Structural Elements". Semester Project. EPFL, 2017.

[15] G. Brekelmans. "Image Segmentation: A Phase Field Approach". BSc Thesis. Eindhoven Univesity of Technology, 2016.

[14] B. Dorussen. "Obtaining Graded Dual Phase Steel Microstructures: An out-of-the-Box Approach". BSc Thesis. Eindhoven Univesity of Technology, 2016.

[13] J. Keulen. "Creating Artificial Microstructures with Phase Field". BSc Thesis. Eindhoven Univesity of Technology, 2016.

[12] J.E.P. van Duuren. "Experimental Characterization of Microstructural Damage and Phase Distribution in Dual Phase Steel". MSc Thesis. Eindhoven Univesity of Technology, 2016.

[11] R. Smeenk. “Phase Field Modeling of Dual-Phase Materials”. BSc Thesis. Eindhoven Univesity of Technology, 2015.

[10] S. Tilmans. “Relating Microstructural Morphology to the Initiation of Fracture: A Monte Carlo Base Approach”. BSc Thesis. Eindhoven Univesity of Technology, 2015.

[9] M. Brands. “Mimicking Microstructures Using a Monte Carlo Based Approach”. BSc Thesis. Eindhoven Univesity of Technology, 2014.

[8] J. Hubregtse. “Identification of Phases in Noise-Polluted Microstructural Images”. BSc Thesis. Eindhoven Univesity of Technology, 2014.

[7] W. Mulder. “Phase Recognition in Dual-Phase Steel Micrographs”. BSc Thesis. Eindhoven Univesity of Technology, 2014.

[6] F. Ramp. “Efficient Computation of Multiple Phase Material Statistics”. BSc Thesis. Eindhoven Univesity of Technology, 2014.

[5] Q. Dronneau. “Establishment of Phase Identification and Damage Software for Microscopic Images of Dual Phase Steel”. Internship. Eindhoven Univesity of Technology, 2013.

[4] P. Hatzidimitris. “Mesh Dependency in Idealized Microstructural Simulations of Dual Phase Steel”. BSc Thesis. Eindhoven Univesity of Technology, 2013.

[3] T. Lapasset. “Characterization of Damage Mechanisms in Dual Phase Steel”. Internship. Eindhoven Univesity of Technology, 2013.

[2] N. Maassen. “Microstructural Modelling of Ductile Failure in Dual-Phase Steel: Increasing the Computational Efficiency”. BSc Thesis. Eindhoven Univesity of Technology, 2013.

[1] J. Ortún. “A Computational Study of Microstructural Geometry Influence in Dual-Phase Steels”. Internship. Eindhoven Univesity of Technology, 2013.

Teaching activities (chronological)

1. Teaching assistant “Statistical Physics 2” (bachelor, EPFL, 2019–2020)
2. Teaching assistant “Continuum Mechanics” (bachelor, EPFL, 2016–2017)
3. Lecturer “Programming project” (bachelor, TU/e, 2011–2015)
4. Teaching assistant “Finite Element Method” (bachelor, TU/e, 2010–2011)
5. Co-developer of a tensor toolbox in Matlab for education purposes (TU/e, 2009–2010)
6. Supervisor “Design Based Learning” (bachelor, TU/e, 2008–2009)

Scientific reviewing activities (journals, alphabetical)

- Computational Materials Science
- Computer Methods in Applied Mechanics and Engineering
- Engineering Fracture Mechanics
- International Journal of Numerical Methods in Engineering
- International of Journal of Fracture
- Journal of the Mechanics and Physics of Solids
- Mechanics of Materials
- Modelling and Simulation in Materials Science and Engineering
- Nature Communications
- Numerical Algorithms
- Physical Review E
- Physical Review Letters
- Strain

Organisation of conferences (chronological)

1. Co-organiser of colloquium of Dr. Elsa Bayart at EPFL (10.2023).
2. Co-organiser of colloquium of Prof.dr. Jan Zeman at EPFL (09.2017).

Institutional responsibilities (chronological)

1. Data champion (EPFL, 2021–pres.).
2. Cluster administrator for the Physics of Complex Systems Laboratory (EPFL, 2017–pres.).
3. Founder of computing cluster support and professionalisation panel (TU/e, 2013–2016).

Outreach activities (chronological)

1. Interview “Horizons” science magizine (FNSF) (2024).
2. Radio interview Swiss National Television (RTS) (2020).
3. Interview by freelance journalist Janny Terlouw (text available on www.geus.me) (2017).
4. Radio interview Rradio (2016).

Programming

- Expert: C++, Python, LaTeX.
- Extensive experience: C, Fortran, OpenMP, Matlab, Bash, Qt.
- Maintainer open-source projects: xtensor, HighFive, python-bibtexparser.
- Developed libraries (selection): GooseFEM, GooseEYE, texplain, ...

Volunteering

- Assistant group leader tour skiing, Alpine Club Switzerland, Lausanne (CH, 2024–pres.).
  Responsibilities: tour planning, assurance of safety.
- Member Commission leisure rowing, Lausanne Sport Aviron, Lausanne (CH, 2019–pres.).
  Responsibilities: group leader outing supervisors leisure rowing, planning and execution travels.
- Outing supervisor leisure rowing, Lausanne Sport Aviron, Lausanne (CH, 2019–pres.).
  Responsibilities: team formation, coaching, assurance of safety.
- President Hora Est, PhD association for (bio-)mechanical engineering (NL, 2014–2015).
  Responsibilities: design and execution social events and career events.
- Buddy of autistic first year student (NL, 2008–2009).
  Responsibilities: enhance well-being and study effectiveness of autistic student.
- Field hockey equipment commissioner, Eindhoven area (NL, 2003–2007).
  Responsibilities: purchase and maintenance of goalkeeper equipment (budget: 8000 euro per year).
- Trainer field hockey, Eindhoven area (NL, 2005–2010).
  Responsibilities: design and execution specialised goalkeeper training.

Other activities

- Tour du Léman (longest row race in the world on closed water) (CH, 2022, 2023).
- Silver Medal Swiss Champion Rowing, Master Men B 4x (CH, 2024, 2024).
- Bronze Medal Swiss Champion Rowing, Master Men B 8+ (CH, 2024).
- Swiss Champion Rowing, Master Men B 4x (CH, 2023).
- Bronze Medal Swiss Championships Rowing, Master Men A 2x (CH, 2021).